

About ForensiQ

A state-of-the-art image forensics platform powered by deep learning and classical signal processing techniques for detecting digital manipulation and AI-generated content.

🔴 Forgery Model

Built on EfficientNetB0 fine-tuned for binary forgery classification. Trained with ImageDataGenerator augmentation including rotation, zoom, flip, and brightness variation on a custom forensics dataset.

📊 Error Level Analysis

ELA detects JPEG re-compression artifacts at different quality levels. Authentic images compress uniformly; forged regions compress at different error levels, creating visible discrepancies our system highlights.

🔍 AI Detection Engine

Uses multi-dimensional heuristics: Gaussian noise profiling, histogram uniformity scoring, and FFT frequency domain analysis. AI-generated images exhibit distinctive spectral signatures in the frequency domain.

🏗️ Architecture Details

EfficientNetB0 → GlobalAveragePooling2D → BatchNorm → Dense(256, ReLU) → BatchNorm → Dropout(0.5) → Dense(128, ReLU) → Dropout(0.5) → Dense(1, Sigmoid). Trained with Adam at lr=1e-4.

📅 Training Pipeline

80/20 train-validation split with early stopping (patience=5), ReduceLROnPlateau (factor=0.3), and ModelCheckpoint saving best val_accuracy weights across 15 epochs max.

⚠️ Limitations

No detection system is 100% accurate. Results should be used as investigative support, not conclusive evidence. AI detection uses heuristics and may misclassify highly compressed or heavily processed real photos.